Predicting Drug–Protein Binding Affinity from Pre-trained Sequence Embeddings

Group #109

Andrew Jing
Awassada Ariyaphuttarat
Hoi Cheung
Sammy Mohamed

1. Background and Motivation

Traditional drug discovery is costly, slow, and limited to wet-lab screening. Bringing a single candidate to clinic can take over a decade and cost billions, with fewer than 10% succeeding in trials. This project asks: can we automate early-stage triage to identify stronger candidates before they reach the bench?

Using transfer learning with the pre-trained transformers ProteinBERT and ChemBERTa, we predict drug-protein binding strength from simple biological and chemical sequences.[1-1, 1-2] This enables a virtual pre-screening of drug-target pairs, helping researchers prioritize high-affinity candidates before costly wet lab testing and accelerating the path to new therapies.

2. Problem Statement

Can independently pretrained transformer embeddings, combined with representation modelling and predictive architectures predict binding accurately enough for drug triage? Does the choice of embedding fusion architecture (MLP, bilinear, cross-attention, or a hybrid) affect performance after tuning?

We frame drug–target interaction (DTI) prediction as a supervised regression task, where the target is the negative log dissociation constant, $pKd = -\log_{10}(Kd)$, where Kd is a binding affinity constant between proteins and drugs, as measured by experimentation. This framing follows the convention established by this seminal paper, a standard benchmark in the field of drug-target interaction. The unit of observation is a single (drug, protein) pair with a measured pKd.[2-1]

3. Data and EDA

The dataset used in this project is the BindingDB,[3-1] accessed from the openly available database TD Commons.[3-2] It consists of (drug, protein, Kd) triples spanning multiple protein and drug families. Targets were annotated with protein-family labels by querying the UniProt REST API for each Target_ID and applying keyword matching over PROTEIN_NAMES and KEYWORDS.[3-3]

As a summary, the following are the most important EDA findings that influenced our modelling choices (see the full EDA in MS4.ipnyb).

1. Assay-ceiling artifact at pKd = 5.0**:** We found a pronounced artifact spike at exactly 5 pKd (Fig 1). This corresponds to 10 μM exactly, a common drug-discovery cutoff that marks low-affinity experiments.[3-4] The 5 pKd reports effectively read as “<5 pKd”, disruptive to regression.
2. Inter-lab measurement noise: Many drug-protein pairs in our dataset have a large number of varying pKd measurements drawn from multiple source publications or repeat experiments.
3. Class imbalance at two scales: at the family level, the corpus is dominated by kinases (~17k samples) and GPCRs (~6,500), with the rarest families having as few as 27 data points (Fig 2); at the per-protein level, the top-20 most-frequent proteins each appear in 150–530 measurements (Fig 3), while the majority of proteins have fewer than 5.

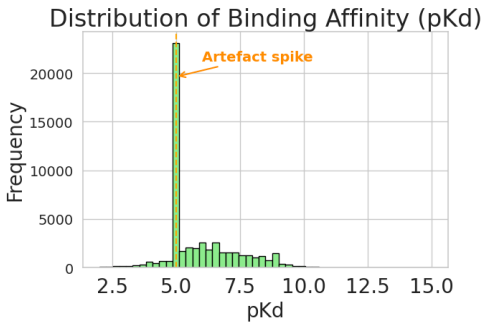


Fig 1: Large artefact spike at pKd=5

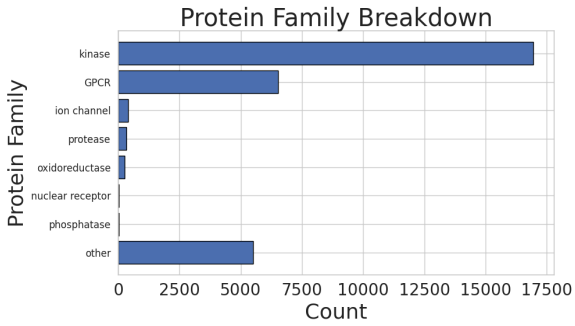


Fig 2: Distribution of proteins per family

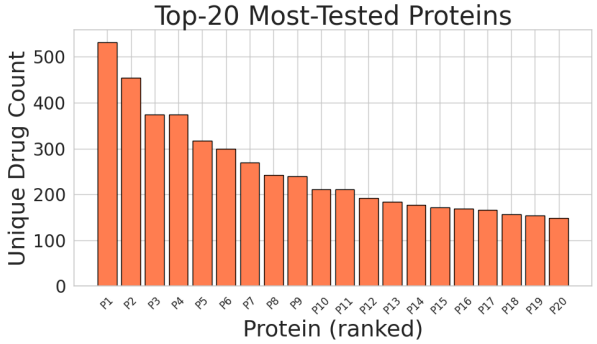


Fig 3: Protein-level drug imbalance

4. Methods

4.1 Data Preprocessing

We first applied standard data hygiene to the raw BindingDB Kd dataset. Exact duplicate rows on (Drug, Target, Y) were dropped, Kd = 0.0 entries were removed, and the remainder was transformed to pKd via $pKd = -\log_{10}(Kd / 10^9)$. Drugs with SMILES exceeding 512 characters were excluded to respect the drug encoder's token limit.[4-1]

To address the non-binding artifact at pKd = 5.0, we dropped all such rows and capped the upper tail at pKd ≤ 12.0; the EDA showed pKd = 5.0 entries correspond to assays reporting their detection threshold rather than measured binding, and values above 12.0 are physically implausible.

To preserve inter-lab measurement spread as a learning signal, we retained all repeat (Drug, Target) measurements rather than aggregating replicates.

To address class imbalance across protein families, we split the data 70/15/15 stratified by family and partitioned by protein, ensuring rare families (some with only 27 examples) appear in all three splits and that no protein leaks across splits. During training we additionally apply a WeightedRandomSampler that up-weights rare proteins.

4.2 Embedding Pipeline

We encode drugs with ChemBERTa-100M-MLM, which tokenizes a SMILES string and produces a representation of shape (sequence length, 768) up to 512 tokens. Proteins are encoded with ProtBERT-BFD, producing a representation of shape (N, 1024), where N is the variable tokenized sequence length and is not subject to any hard cap.

The first position of each output is a CLS token pre-trained as a pooled summary of the full sequence; the MLP and bilinear models consume only the CLS embeddings, while the cross-attention and hybrid models use the full token sequences. All embeddings are pre-computed and cached, equivalent to freezing the encoders end-to-end.

4.3 Common training protocol across all experiments

All experiments share a common training protocol. Models were trained with MSE loss on pKd using the Adam[4-2] or AdamW[4-3] optimizers and gradient clipping at max-norm 1.0 to stabilize training. We used early stopping on validation loss and retained the best-validation checkpoints. To address per-family class imbalance, we applied a WeightedRandomSampler drawing each example with probability proportional to $1/\sqrt{(\text{per-protein count})}$. This up-weights rare proteins without extreme over-sampling of singletons.

4.4 Fusion-head Architectures

4.4.1 MLP Model

The MLP serves as our simplest fusion baseline. By concatenating the drug and protein CLS representations and passing them through a feed-forward network, the model makes no assumption about how the 2 modalities should interact, leaving the fusion function to be learned just from the data, and therefore our reference model.

The model takes the drug and protein CLS embeddings (768 and 1024 dimensions), applies LayerNorm to each independently, and concatenates them into a 1792-dimensional vector. This is passed through two hidden blocks of widths 512 and 256 (Linear → BatchNorm1d → ReLU → Dropout) and a final linear layer maps the results to a scalar pKd (Fig 4).

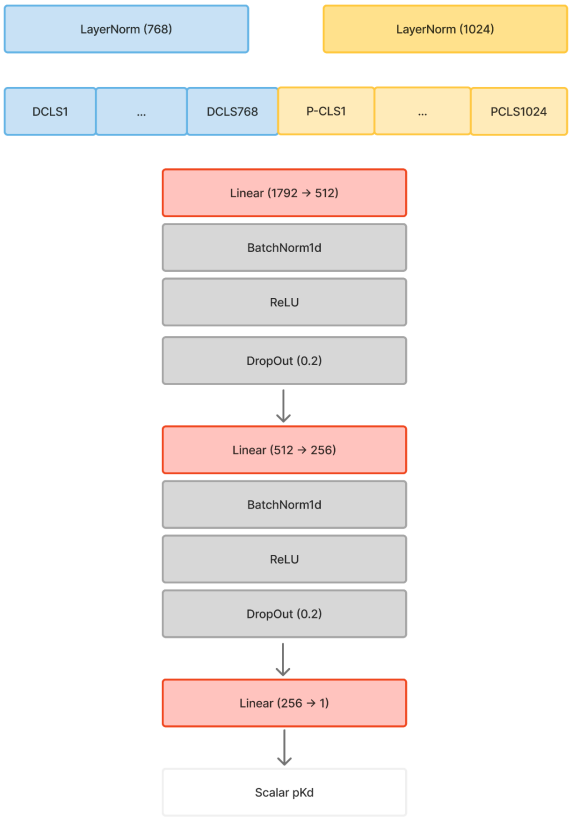


Fig 4: Architecture of MLP Model

4.4.2 Bilinear Model

Binding affinity is fundamentally a function of how specific structural features of the drug and protein molecules interact with each other. Rather than relying on unstructured non-linearities in the MLP head to approximate these interactions, the logical next step is the bilinear model, which learns global attributes of the proteins and drugs, and then interacts between those attributes.

The model operates on the same CLS embeddings as the MLP (Fig 5). Each modality is Layer-Normalized and projected into a learned 256-dimensional space via a Linear → ReLU block. Then, the drug and protein representations are combined into 256 channels capturing global interactions via a bilinear fusion layer.

The output vector of the bilinear fusion is y, with dims (256,) for the 256 interaction channels. Each component in y is a learned interaction across drug and protein ($y_k = d^T W_k p + b_k$), and as such we also have 256 weight-matrices W_k each with dims (256,256), as well as a total of 256 biases b_k. So, each element of the output vector is a weighted combination of learned global features of the drug and protein, allowing a more nuanced interaction between drug and protein than the MLP.

Finally, the interaction vector passes through a BatchNorm1d normalization followed by ReLU and Dropout (p=0.2) before a final layer maps to a scalar pKd prediction.

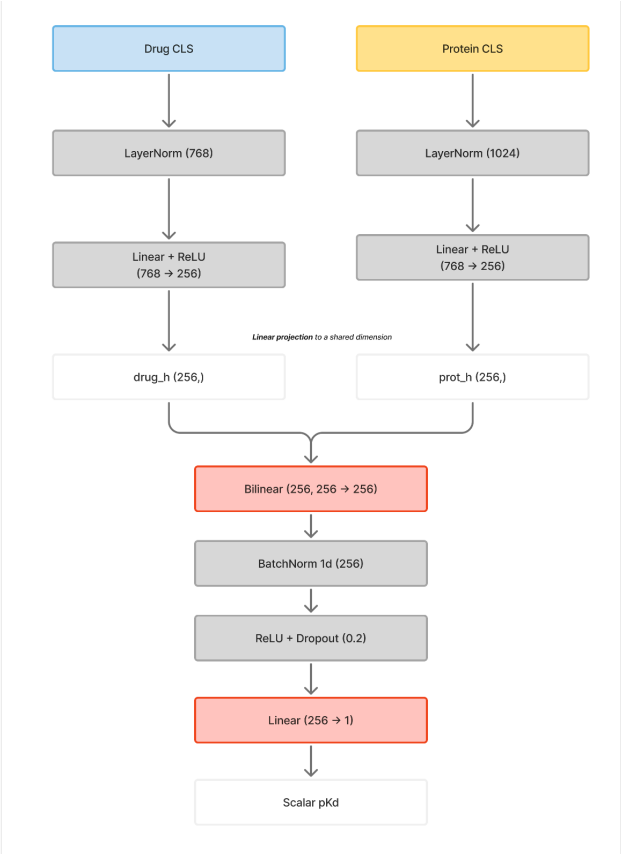
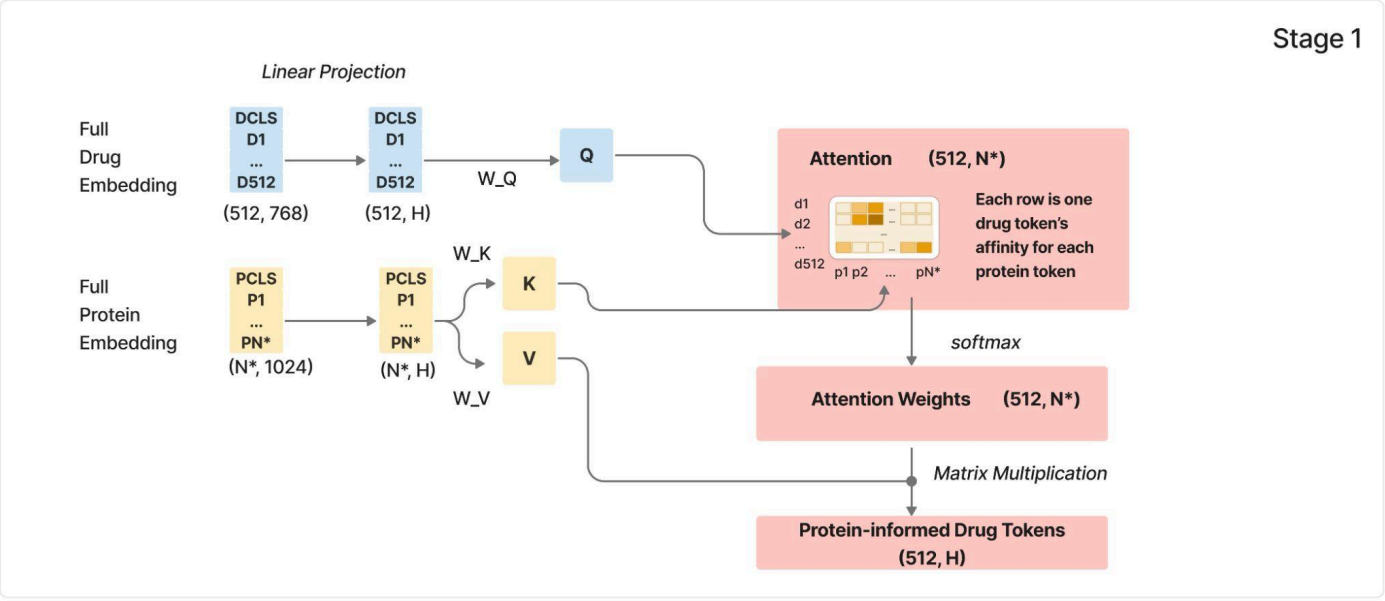


Fig 5: Architecture of Bilinear Model

4.4.3 Cross Attention

The MLP and bilinear heads operate on pooled CLS representations, discarding per-token resolution of the underlying embeddings. However, binding is governed by structural contacts between specific drug structures and protein residues. A natural inductive bias is cross-attention, which allows the model to capture this relationship by attending across the drug and protein token sequences in both directions: the drug queries the protein, and the protein queries the drug.

Both sequences are first linearly projected into a shared hidden dimension H=128, yielding a (512, H) drug representation and an (N, H) protein representation (Fig 6, Stage 1). We then apply two cross attention blocks in sequence. In the first block, drug tokens serve as queries and protein tokens as keys and values, producing protein-informed drug tokens via the standard scaled dot-product attention $\text{softmax}(QK^T/\sqrt{H})V$ (Fig 6, Stage 1). In the second block, those protein-informed drug tokens become the keys and values, while protein tokens serve as queries, yielding drug-informed protein tokens (Fig 6, Stage 2). Each set of attended tokens is then mean-pooled along its sequence dimension, the two resulting (H,) vectors are concatenated, and the combined 2H-dimensional representation is passed through a small MLP head that produces the scalar pKd prediction (Fig 6, Stage 3).



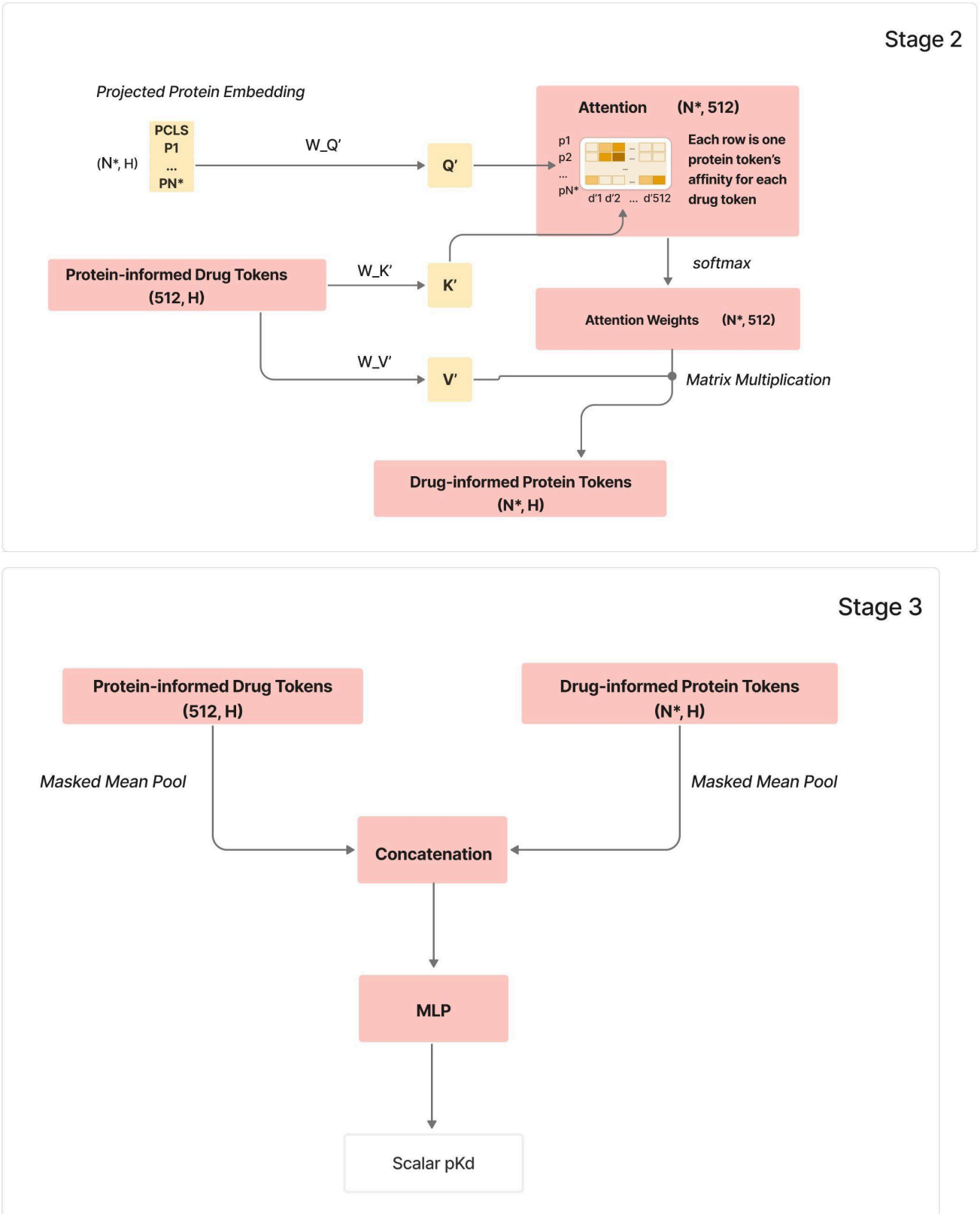


Figure 6: Architecture of the cross-attention model.

4.4.4 Hybrid Architecture

The bilinear head captures global multiplicative structure between pooled modality vectors; cross-attention captures local, position-specific interactions over full token sequences. These inductive biases are complementary, and binding can plausibly depend on both. The hybrid runs the two pathways in parallel and concatenates their outputs, testing whether they contribute independent signals.

The model (Fig 7) takes both CLS embeddings (for the bilinear pathway) and full token sequences (for the cross-attention pathway), with small additive Gaussian noise (σ = 0.01) on the inputs as regularization. Representations are projected into two hidden spaces with dimension sizes of 64 and 128 for cross-attention and bilinear respectively. As before, the cross-attention pathway uses 4-head attention applied drug-to-protein then protein-to-drug, followed by masked-mean pooling, yielding a 128-dimensional vector; the bilinear pathway produces a 128-dimensional vector of global feature interactions. The two outputs are LayerNorm-normalized independently, concatenated to 256 dimensions, and passed through a feed-forward head (256 → 128 → 1) to produce the scalar pKd prediction.

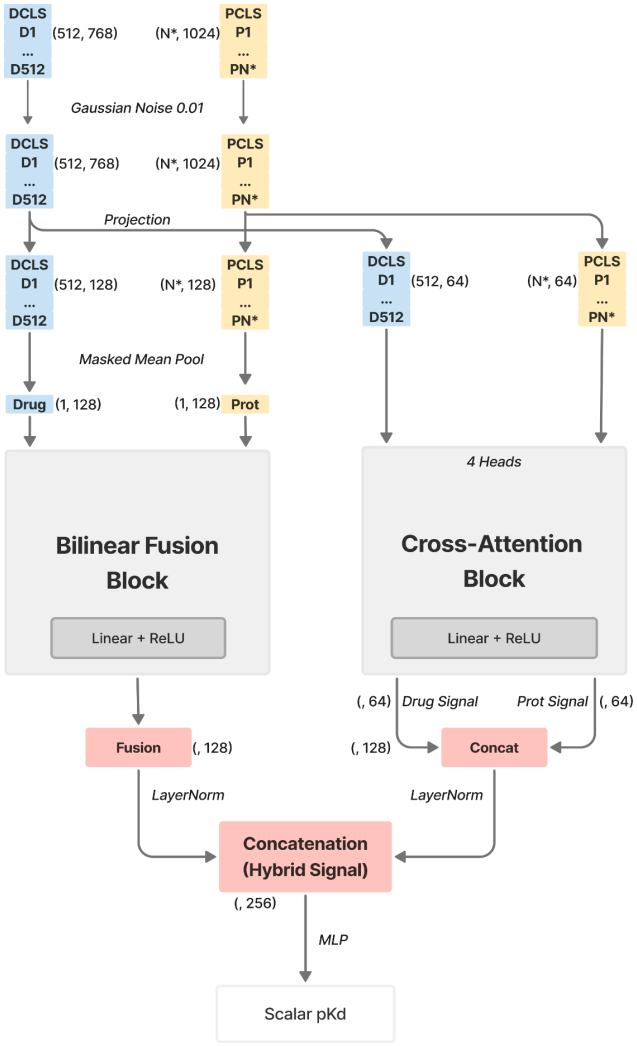


Fig 7: Architecture of hybrid model (N*: Longest protein length)

4.5 Hyperparameter search

Across the baselines, the dominant failure mode was overfitting. We therefore ran 4 hyperparameter search arms, one per architecture, sharing a common regularization-focused agenda and adding model-specific changes. Searches for the MLP and bilinear models were performed using an agent-based automated loop;[4-4] the cross-attention search used Optuna's Bayesian search with pruning.[4-5] The hybrid search was conducted manually due to the model's ~1 hour-per-experiment compute cost making automated sweeps infeasible. Table 8 summarizes the search space per architecture.

Category	MLP	Bilinear	Cross Attention	Hybrid
Architecture	hidden dims, layer ordering, norm type (BN vs LN)	hidden_dim (128, 256, 512), low-rank factorization (rank=64), deeper head, activation	hidden_dim (64, 128, 256)	num heads, pathway hidden dims, add vs concat, ReLU vs GELU
Regularization	dropout, weight decay, label noise, modality dropout	dropout (0.1–0.5), weight decay (1e-4), label noise (σ=0.05), modality dropout (p=0.1)	dropout (0.1-0.3), weight decay (1e-5 to 1e-3), pathway dropout (10–20%)	dropout, pathway dropout, input Gaussian noise, oversampling, weight decay
Activations	ReLU, GELU, SiLU, SwiGLU, ELU	ReLU vs SiLU	-	ReLU vs GELU
Training regime	LR (with cosine annealing), batch size	LR (1e-3, 5e-4, 1e-4), cosine annealing, batch size (128, 256), extended epochs (early stopping → fixed 80-epoch budget)	-	LR, scheduler strategy, batch size
Data augmentation	Mixup (multiple α), manifold Mixup, Gaussian noise on embeddings /labels	-	-	-
Search method	Agent-based (~109 runs)	Agent-based (23 runs)	Optuna Bayesian	Manual (~1 hr/run)

Table 8. Hyperparameter search space per architecture. '-' indicates a category that was not explored for that model. Specific values appear where they were swept over a discrete set. Otherwise, ranges or qualitative categories are given.

5. Results and Interpretation

5.1 Comparison Across Fusion Heads

Table 9 reports the baseline and best post-finetuning RMSEs on train, validation and test sets for each of the four fusion heads, alongside a naive baseline that predicts the mean pKd value.

The headline finding of our experiments is that test performance was inversely correlated with model complexity. The MLP, our simplest fusion head, achieved the best test RMSE (1.30), followed by the bilinear model (1.311), the cross-attention model (1.355), and the hybrid (1.453). All four models still sit approximately 2-3 times the inter-lab noise floor of 0.4-0.6 pKd units identified in the EDA, indicating that none of them has saturated the achievable performance on this regressive task. The fact that the four architectures cluster within a narrow band, despite very different inductive biases, suggests that further gains will likely come from changes to the encoders or training data rather than from richer fusion heads as we will discuss in the Discussion section.

	Train RMSE		Validation RMSE		Test RMSE	
	Baseline (MS3)	Tuned (MS4)	Baseline (MS3)	Tuned (MS4)	Baseline (MS3)	Tuned (MS4)
Naive	1.5364	-	1.5627	-	1.4850	-
MLP	0.7058	**0.7793**	1.3577	**1.2527**	1.3851	**1.3030**
Bilinear	0.7403	1.2959	1.4206	1.4024	1.3313	1.3217
Cross Attention	1.0109	0.9828	1.3364	1.3504	1.4787	1.3550
Hybrid Model	-	1.1017	-	1.3248	-	1.4526

Table 9. RMSE on training, validation, and test sets for each fusion architecture, comparing the MS3 baseline configuration to the best configuration found through hyperparameter search. The naive model universally predicts the training set's mean pKd. Lower is better; bold indicates the best tuned RMSE across architectures.

5.2 Hyperparameter Search Results

5.2.1 MLP Model

The search identified three complementary changes that reduced validation RMSE from 1.36 to 1.23.[1]

The single most impactful change was modality dropout (p = 0.10), which randomly zeros out either the drug or protein embedding for each sample independently during training, forcing the model to predict from partial inputs. This alone improved the RMSE by 6%. Combining it with a wider architecture ([1024, 512, 256]) and Mixup Augmentation[5-1] (α = 0.4) produced the remaining improvement.

Alternative activations (GELU, SiLU, SwiGLU, ELU) consistently underperformed plain ReLU, likely because the dataset is small and the embeddings are already well-conditioned. Both Gaussian noise on embeddings and labels worsened validation RMSE and were dropped.

Test RMSE improved across all protein families except nuclear receptors (Fig 10), the least-represented family. Additional regularization likely reduced its already limited effective training signal.

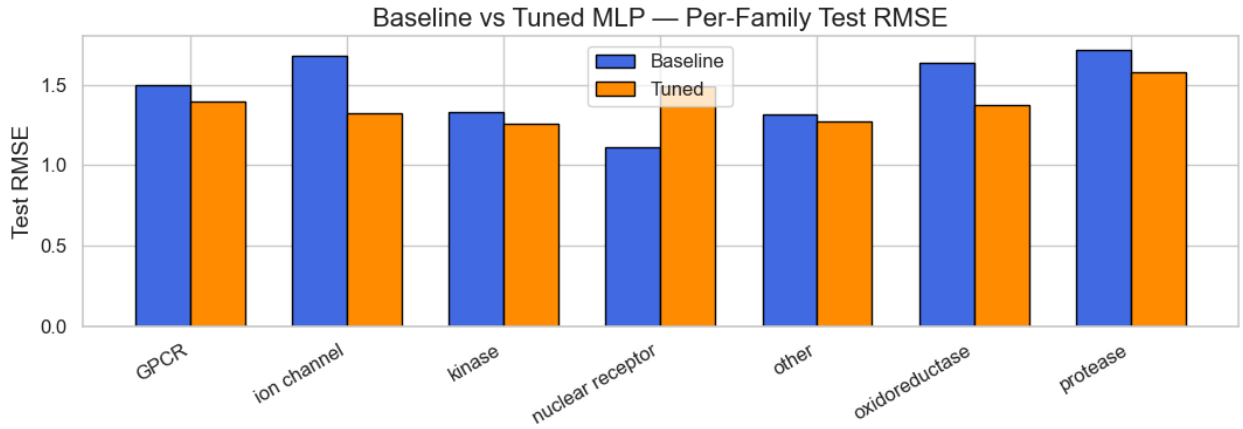


Figure 10: Per family test RMSE comparison between MLP baseline and fine-tuned model.

[1] Reproducing the configuration in the final notebook yielded a slightly higher RMSE of 1.25 (shown in the comparison table) due to training non-determinism despite using the same seed. The results reported in the main text are based on the search.

5.2.2 Bilinear Model

The search reduced validation RMSE from 1.42 to 1.40 through three changes. Increasing dropout from p=0.2 to p=0.4 was the dominant gain. The bilinear weight tensor (256×256×256) has enormous capacity and is prone to memorizing training patterns. Replacing BatchNorm1d with LayerNorm and increasing dropout to p=0.4 in the improved stability and reduced overfitting. Low-rank bilinear factorization (rank=64) was explored as a compressed alternative. While it's 60× faster, it cost 0.07 RMSE units and was not retained.

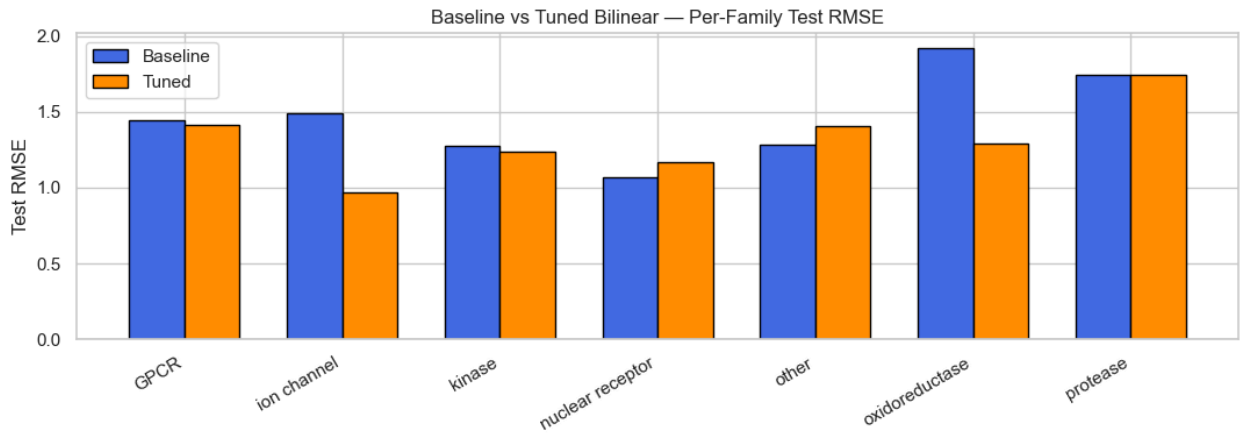


Figure 11: Per family test RMSE comparison between Bilinear baseline and fine-tuned model.

5.2.3 Cross-Attention

The search yielded values similar to those from the initial configuration in Milestone 3. The changes were an increase in hidden dimension from 32 to 128, and a probability of 0.1 of skipping an attention block. This produced a less overfit model that performed better on the test set than before. The per-family test RMSE was improved for almost all protein families.

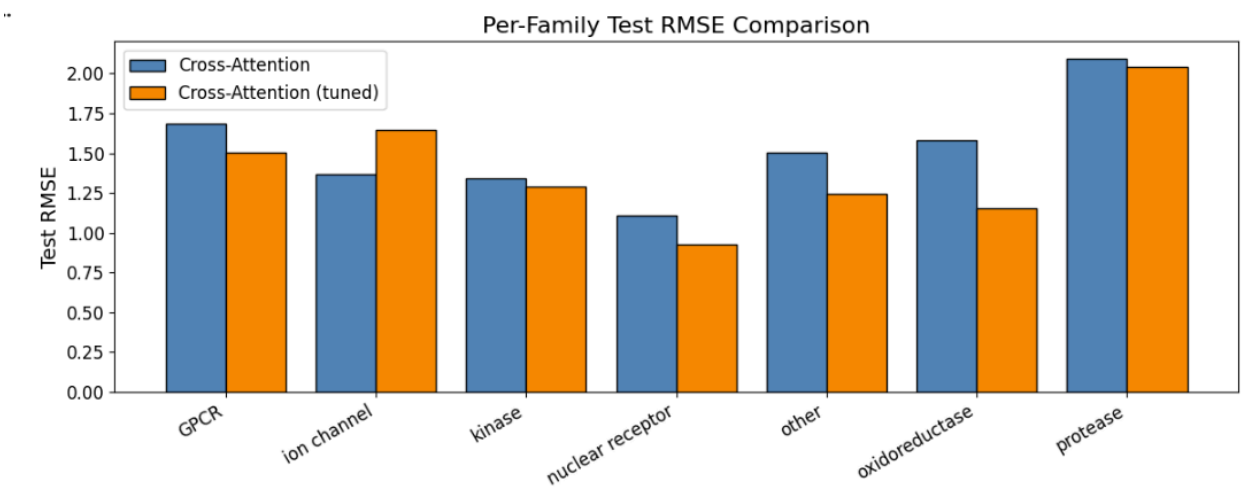


Figure 12: Per family test RMSE comparison between cross-attention baseline and fine-tuned model.

5.2.4 Hybrid

The hybrid model was highly unstable under hyperparameter tuning, and computational cost (~1 hr per experiment) made automated search infeasible. Tuning was therefore manual and coarse, but two hyperparameters stood out.

Batch size produced the single largest improvement. The initial setup (batch size 256, high LR) plateaued at validation RMSE 1.46; reducing batch size to 8 with a correspondingly lower LR brought validation RMSE down to 1.32.

Pathway dropout, in which at most one of the cross-attention or bilinear branches is zeroed per batch and the surviving branch is scaled by 2 (analogous to standard dropout), was the primary regularization win. At p = 0.15 (70% both pathways active, 15% cross-attention dropped, 15% bilinear dropped), the validation-to-training loss ratio fell from 5:1 to 2:1.

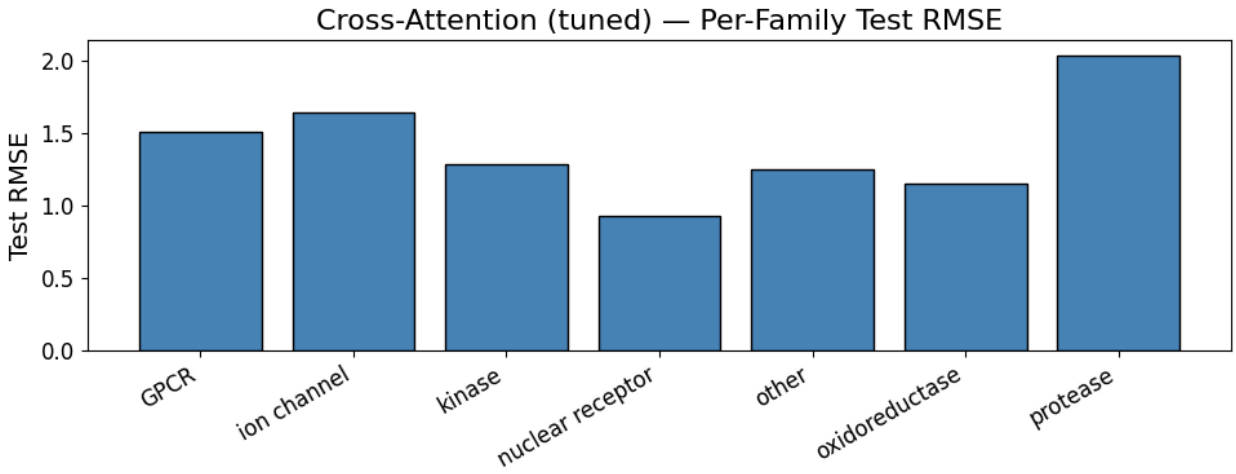


Figure 13: Per family test RMSE of hybrid model (no MS3 baseline available for comparison).

6. Conclusion and Discussion

6.1 Main findings and interpretation

The most striking result of this work is the inverse relationship between model complexity and test performance: the MLP, our simplest fusion head, achieved the lowest test RMSE, while the hybrid model performed worst. Neither the bilinear or cross-attention mechanism offered advantages over the MLP despite their architectural priors for multiplicative and position-specific interactions.

The plateau across architectures most plausibly reflects two upstream limitations: neither encoder was pre-trained with a binding-affinity objective, and sequence-only representations cannot capture the three-dimensional geometry and electrostatics that govern binding in practice. More expressive fusion heads cannot recover information the encoders never extracted.

Another pattern across all four searches was that the most impactful interventions were uniformly regularization-related. This suggests that, given the current data scale and embedding choice, the binding constraint on test performance is generalization rather than capacity.

6.2 Limitations

The most significant limitation is the use of frozen encoders, which almost certainly limits achievable test RMSE for the reasons discussed above. We had intended to fine-tune ProtBERT-BFD and ChemBERTa-100M-MLM end-to-end on Kd, but underestimated the GPU memory required to backpropagate through both encoders simultaneously and lacked the compute to proceed.

A second limitation is protein diversity. Although the dataset contains tens of thousands of (drug, protein) pairs, these span only ~1,000 unique proteins in training and ~200 each in validation and test. Combined with per-protein partitioning, this leaves the held-out splits statistically noisy and explains the validation-to-test gap visible for several architectures.

The third limitation is the inter-lab measurement noise in BindingDB labels (~0.4-0.6 pKd standard deviation). This makes model learning difficult and sets a hard floor on the achievable test RMSE.

6.3 Future work

The most direct extension to the project is to fine-tune the encoders, particularly with parameter-efficient methods (LoRA,[6-1] adapter layers) that fit within modest GPU budgets. Our results also highlight the need to source a larger corpus of data, particularly one with broader protein diversity. A complementary direction is to incorporate three-dimensional structural information from open sources such as AlphaFold,[6-2] since sequence-only encoders cannot represent the geometry and electrostatics that ultimately govern binding. In terms of training, we should also cross-pollinate the most impactful hyperparameter and architecture changes across models. For example, using modal dropout for the hybrid model or across the bilinear latent

spaces, since it was the dominant gain in the MLP search but was not tested in the other architectures.

7. Broader Impact

The contribution of this work is methodological and diagnostic rather than applied. Our best test RMSE of 1.30 corresponds to a roughly 20-fold error on the Kd scale, too imprecise for deployment as a triage tool. What this project offers instead is empirical evidence about which regularization and architecture did or did not help within a modest compute budget, which is particularly relevant for under-sourced research settings. Bottlenecks identified here also remain relevant for any future modeling work in this space: the protein-family skew in standard binding datasets means screening capacity will continue to be biased toward already well-studied targets unless dataset composition itself improves, and even a perfectly accurate Kd predictor would address only one early step in a much longer pipeline.

8. References

1-1: ProteinBERT

Brandes, N., Ofer, D., Peleg, Y., Rappoport, N., & Linial, M. (2022). ProteinBERT: A universal deep-learning model of protein sequence and function. *Bioinformatics*, *38*(8), 2102–2110. https://doi.org/10.1093/bioinformatics/btac020

1-2: ChemBERTa

Chithrananda, S., Grand, G., & Ramsundar, B. (2020). *ChemBERTa: Large-Scale Self-Supervised Pretraining for Molecular Property Prediction* (arXiv:2010.09885). arXiv. https://doi.org/10.48550/arXiv.2010.09885

2-1: Motivational paper

Öztürk, H., Özgür, A., & Ozkirimli, E. (2018). DeepDTA: Deep drug–target binding affinity prediction. *Bioinformatics*, *34*(17), i821–i829. https://doi.org/10.1093/bioinformatics/bty593

3-1: Dataset

Liu, T., Lin, Y., Wen, X., Jorissen, R. N., & Gilson, M. K. (2007). BindingDB: A web-accessible database of experimentally determined protein-ligand binding affinities. *Nucleic Acids Research*, *35*(Database issue), D198-201. https://doi.org/10.1093/nar/gkl999

3-2: Database

Drug-Target Interaction. (n.d.). TDC. Retrieved May 12, 2026, from http://localhost:4000/multi_pred_tasks/dti/

3-3: Protein Families

UniProt Consortium. (2023). UniProt: The Universal Protein Knowledgebase in 2023. *Nucleic Acids Research*, *51*(D1), D523–D531. https://doi.org/10.1093/nar/gkac1052

3-4: Measurement artifact

Determination of Ligand-Binding Affinity (K d) Using Transverse Relaxation Rate (R 2) in the Ligand-Observed 1H NMR Experiment and Applications to Fragment-Based Drug Discovery. (2023). *Journal of Medicinal Chemistry*, *66*(15), 10617–10627. https://doi.org/10.1021/acs.jmedchem.3c00758

Irwin, J. J., Duan, D., Torosyan, H., Doak, A. K., Ziebart, K. T., Sterling, T., Tumanian, G., & Shoichet, B. K. (2015). An Aggregation Advisor for Ligand Discovery. *Journal of Medicinal Chemistry*, *58*(17), 7076–7087. https://doi.org/10.1021/acs.jmedchem.5b01105

What Kd cut-off values are the accepted for consider a protein-ligand complex with a low, medium or high affinity? (n.d.). ResearchGate. Retrieved May 12, 2026, from https://www.researchgate.net/post/What-Kd-cut-off-values-are-the-accepted-for-consider-a-protein-ligand-complex-with-a-low-medium-or-high-affinity

Bosc, N., Meyer, C., & Bonnet, P. (2017). The use of novel selectivity metrics in kinase research. *BMC Bioinformatics*, *18*, 17. https://doi.org/10.1186/s12859-016-1413-y

4-1: SMILES Tokenization

Weininger, D. (1988). SMILES, a chemical language and information system. 1. Introduction to methodology and encoding rules. *Journal of Chemical Information and*

Computer Sciences, *28*(1), 31–36. https://doi.org/10.1021/ci00057a005

4-2: Adam Optimizer
Kingma, D. P., & Ba, J. (2017). *Adam: A Method for Stochastic Optimization* (arXiv:1412.6980). arXiv. https://doi.org/10.48550/arXiv.1412.6980

4-3: AdamW Optimizer
Loshchilov, I., & Hutter, F. (2019). *Decoupled Weight Decay Regularization* (arXiv:1711.05101). arXiv. https://doi.org/10.48550/arXiv.1711.05101

4-4: Autoresearch Repository
Cheung, H. C. (2026). *Hoi-Ching-Cheung/autoresearch_mlp* [Python]. https://github.com/Hoi-Ching-Cheung/autoresearch_mlp (Original work published 2026)

4-5: Optuna Bayesian Search
Akiba, T., Sano, S., Yanase, T., Ohta, T., & Koyama, M. (2019). *Optuna: A Next-generation Hyperparameter Optimization Framework* (arXiv:1907.10902). arXiv. https://doi.org/10.48550/arXiv.1907.10902

5-1: Mixup Augmentation
Zhang, H., Cisse, M., Dauphin, Y. N., & Lopez-Paz, D. (2018). *mixup: Beyond Empirical Risk Minimization* (arXiv:1710.09412). arXiv. https://doi.org/10.48550/arXiv.1710.09412

6-1: Low-Rank Adaptation
Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., & Chen, W. (2021). *LoRA: Low-Rank Adaptation of Large Language Models* (arXiv:2106.09685). arXiv. https://doi.org/10.48550/arXiv.2106.09685

6-2: AlphaFold Prediction
Jumper, J., Evans, R., Pritzel, A., Green, T., Figurnov, M., Ronneberger, O., Tunyasuvunakool, K., Bates, R., Žídek, A., Potapenko, A., Bridgland, A., Meyer, C., Kohl, S. A. A., Ballard, A. J., Cowie, A., Romera-Paredes, B., Nikolov, S., Jain, R., Adler, J., … Hassabis, D. (2021). Highly accurate protein structure prediction with AlphaFold. *Nature*, *596*(7873), 583–589. https://doi.org/10.1038/s41586-021-03819-2

Jupyter Notebook Implementation:

This project used Claude and ChatGPT to support some implementation details of our described models. During model tuning, regulatory architecture changes used generative AI to integrate the team's changes into existing infrastructure.